UNIVERSITY AT ALBANY
State University of New York

College of Arts and Sciences
Department of Mathematics and Statistics

Algebra/Topology Seminar

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Density-Sensitive Bifiltrations in Topological Data Analysis

Thursday, November 5, 2026
3:00 p.m. in Massry B008

Abstract. The standard filtrations of TDA (Rips, Čech, Delaunay) are stable to small perturbations of the input data, but are notoriously unstable to outliers, and can be insensitive to variations in density. One of the main motivations of 2-parameter persistence is to address these limitations. There are several natural ways one can define a density-sensitive bifiltration of point cloud or metric data, offering different tradeoffs between generality, computability, and robustness to outliers. Such bifiltrations have been actively studied in recent years, leading to substantial advances in our understanding of them, as well as new computational tools. In this talk, I will survey this progress.